

CMOS Inverter Design and Characterization

Using CMOS SCL 180 nm Technology

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1 Introduction and Design Specifications

A CMOS inverter is the fundamental building block of digital CMOS logic circuits. It consists of a PMOS transistor connected between V_{DD} and the output node, and an NMOS transistor connected between the output node and ground, with both gates tied to a common input, as shown conceptually in Figure ??.

Table 1: Design Specifications

Parameter	Value
Technology	CMOS SCL 180 nm
Supply voltage, V_{DD}	1.8 V
Channel length (NMOS & PMOS), L	$0.18\ \mu\text{m}$ (minimum)
Load capacitance, C_L	30 fF
Simulator	Cadence Spectre (Virtuoso ADE-L)

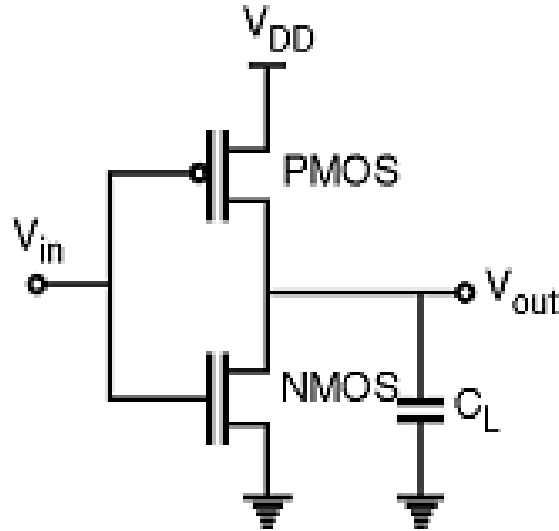


Figure 1: Conceptual schematic of a CMOS inverter with output load capacitor C_L .

2 Part A

2.1 Q1: CMOS Inverter Schematic and Sizing (W/L Ratios)

2.1.1 Theory

For a first-order (long-channel, square-law) sizing approach, the PMOS device is sized wider than the NMOS device to compensate for the lower hole mobility (μ_p) compared to electron mobility (μ_n). The drive strength of a MOSFET in saturation is proportional to

$$I_D \propto \mu C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2.$$

To obtain matched pull-up and pull-down strengths (and hence a switching threshold close to $V_{DD}/2$), the PMOS width is ideally chosen as

$$\left(\frac{W}{L}\right)_p = \frac{\mu_n}{\mu_p} \left(\frac{W}{L}\right)_n.$$

As a starting point for the initial design, however, both the NMOS and PMOS were sized identically at the minimum device width available in the 180 nm node, $W_n = W_p = 0.42 \mu\text{m}$, with the minimum channel length $L_{min} = 0.18 \mu\text{m}$ for both devices. As shown in Q2, this equal-width starting point does *not* give a symmetric VTC ($V_M \neq V_{DD}/2$), and the PMOS is subsequently widened in Q4 to correct this.

2.1.2 Final Sizing Used

Table 2: Transistor sizing used for the initial (Part A, Q1–Q3) design

Device	W (μm)	L (μm)	W/L
NMOS (M0)	0.42	0.18	2.33
PMOS (M1)	0.42	0.18	2.33

$$\left(\frac{W}{L}\right)_p / \left(\frac{W}{L}\right)_n = \frac{2.33}{2.33} = 1.00$$

With equal W/L ratios, the PMOS pull-up network is (given $\mu_p < \mu_n$) weaker than the NMOS pull-down network, which is expected to skew the VTC — this is examined quantitatively in Q2. The load capacitor was set to $C_L = 30 \text{ fF}$ as specified.

2.1.3 Schematic

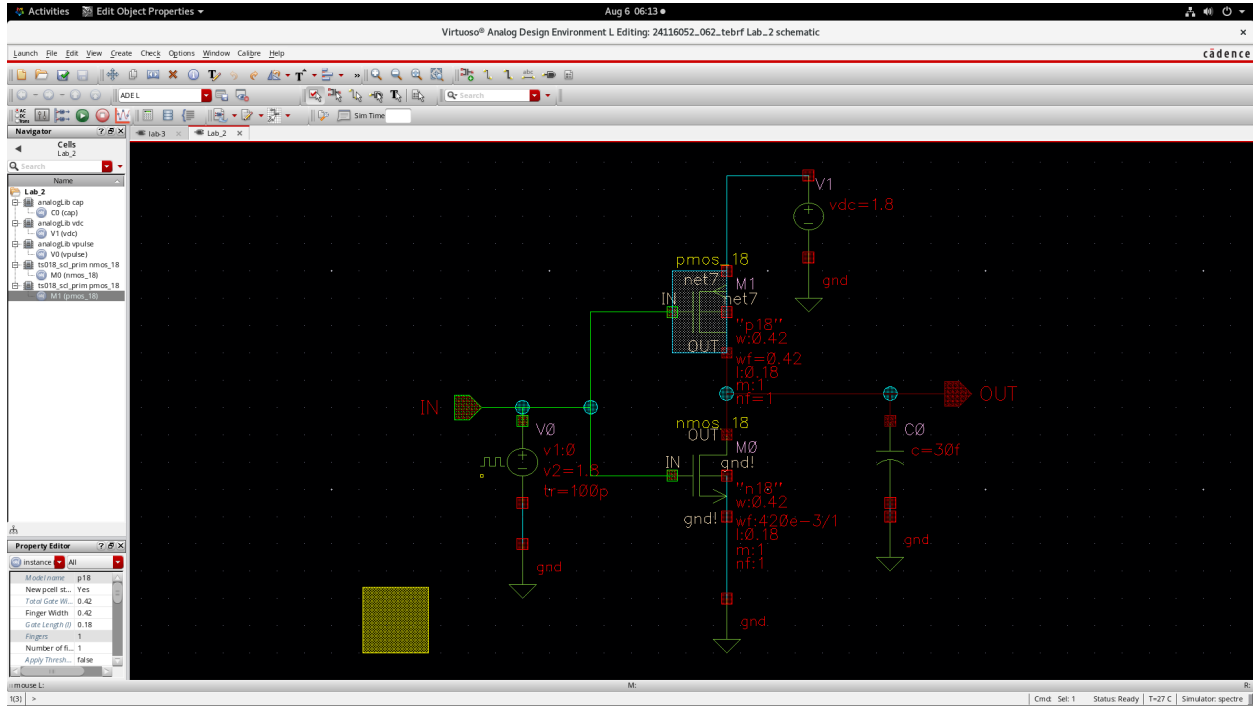


Figure 2: CMOS inverter schematic in Cadence Virtuoso showing the initial sizing: NMOS (M0: $W/L = 0.42/0.18$), PMOS (M1: $W/L = 0.42/0.18$), pulse source V_0 (0 to 1.8 V) at the input, DC supply $V_1 = 1.8$ V, and load capacitor $C_0 = 30$ fF at the output.

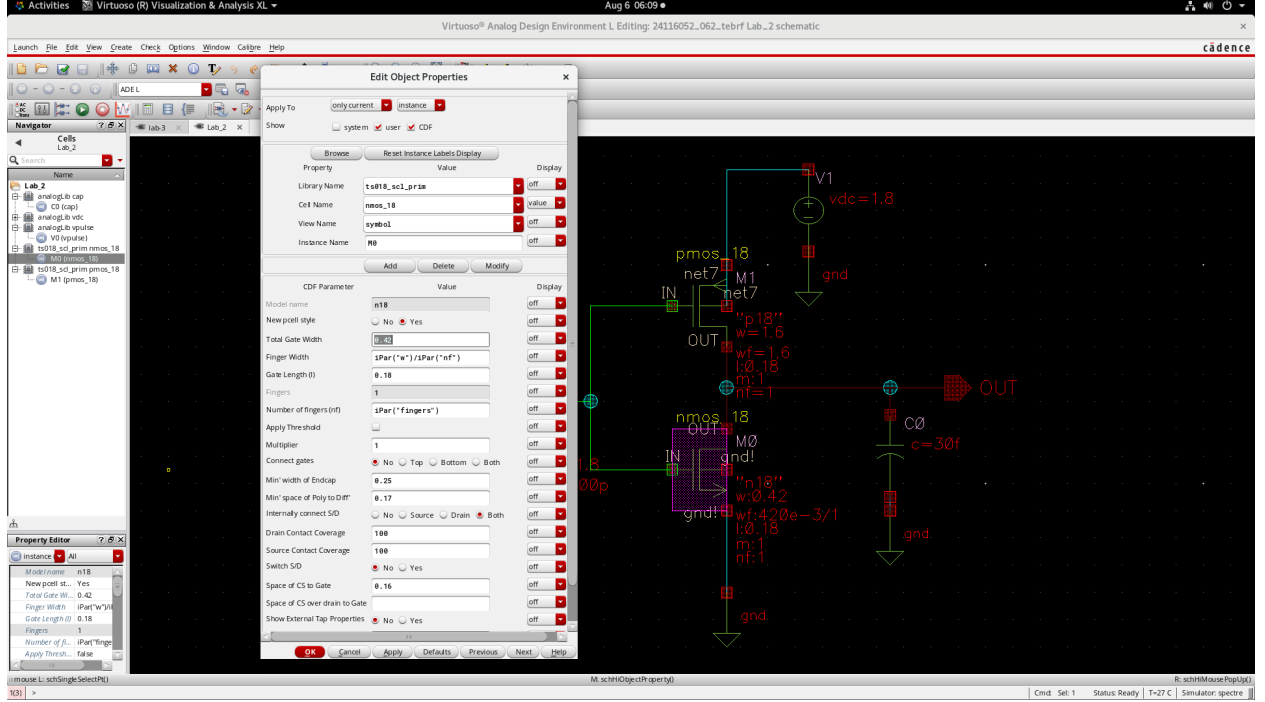


Figure 3: NMOS (M0) CDF parameters: $W = 0.42 \mu\text{m}$, $L = 0.18 \mu\text{m}$, fingers = 1. (This sizing is used for the NMOS in both the initial and the modified design — the NMOS is never changed.)

2.2 Q2: DC Voltage Transfer Characteristics (VTC) and Switching Threshold

2.2.1 Theory

The DC voltage transfer characteristic (VTC) is obtained by sweeping the input DC voltage V_{in} from 0 to V_{DD} and plotting the resulting output voltage V_{out} . The switching threshold voltage V_M is defined as the input voltage at which $V_{out} = V_{in}$, i.e., the point where the VTC curve intersects the line $V_{out} = V_{in}$ (unity-gain point). At this point, both the NMOS and PMOS transistors are in saturation and conduct the same current.

Using the simplified long-channel (alpha-power law, $\alpha = 2$) model and neglecting channel-length modulation, V_M can be approximated as

$$V_M = \frac{V_{DD} - |V_{TP}| + V_{TN} \sqrt{\frac{k_n}{k_p}}}{1 + \sqrt{\frac{k_n}{k_p}}}, \quad k_{n,p} = \mu_{n,p} C_{ox} \left(\frac{W}{L} \right)_{n,p}. \quad (1)$$

For a symmetric design where $k_n = k_p$ (i.e., the pull-up and pull-down strengths are matched) and $V_{TN} \approx |V_{TP}|$, Equation 1 reduces to $V_M \approx V_{DD}/2$.

Also plotted is the supply current I_{DD} (current drawn from V_{DD}) versus V_{in} , which shows the well-known short-circuit / crowbar current peak near $V_{in} = V_M$, where both transistors are simultaneously ON.

2.2.2 Simulation Setup

A dc analysis was run in Spectre, sweeping V_{in} (the DC voltage source at the gate) from 0 V to 1.8 V, and plotting V_{out} (red trace) against V_{in} (green trace, plotted as the diagonal reference line $V_{out} = V_{in}$).

2.2.3 Results



Figure 4: Simulated DC Voltage Transfer Characteristic (VTC) of the initial inverter design: V_{out} (red) vs. V_{in} (green, diagonal reference). The steep transition region is centered close to $V_{in} \approx 0.74$ V.

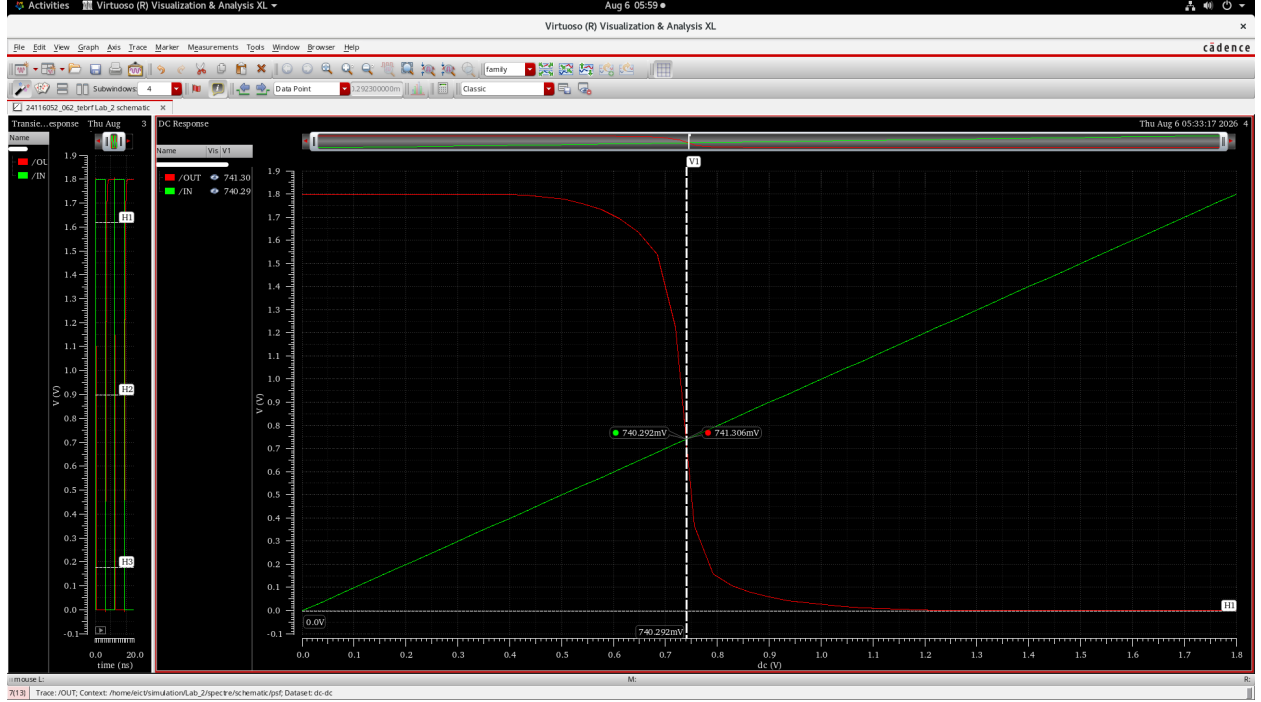


Figure 5: Marker placed at the intersection of V_{out} and $V_{in} = V_{out}$ line, giving the switching threshold $V_M \approx 740.29 \text{ mV (IN)} / 741.31 \text{ mV (OUT)} \Rightarrow V_M \approx 0.740 \text{ V}$.

Table 3: Switching threshold voltage extracted from simulation

Quantity	Value
Switching threshold, V_M (simulated)	$\approx 0.740 \text{ V}$
V_M/V_{DD}	0.411

Observation: Since $V_M = 0.740 \text{ V} < V_{DD}/2 = 0.9 \text{ V}$, the VTC transition is skewed towards the left (lower V_{in}). This is expected: with $W_n = W_p = 0.42 \mu\text{m}$ (equal W/L for both devices), the lower hole mobility μ_p relative to electron mobility μ_n makes the PMOS pull-up network comparatively *weaker* than the NMOS pull-down network. A weaker PMOS means the output starts falling (i.e., the NMOS starts winning) at a lower input voltage, which pulls V_M below $V_{DD}/2$. This is corrected in Q4 by widening the PMOS.

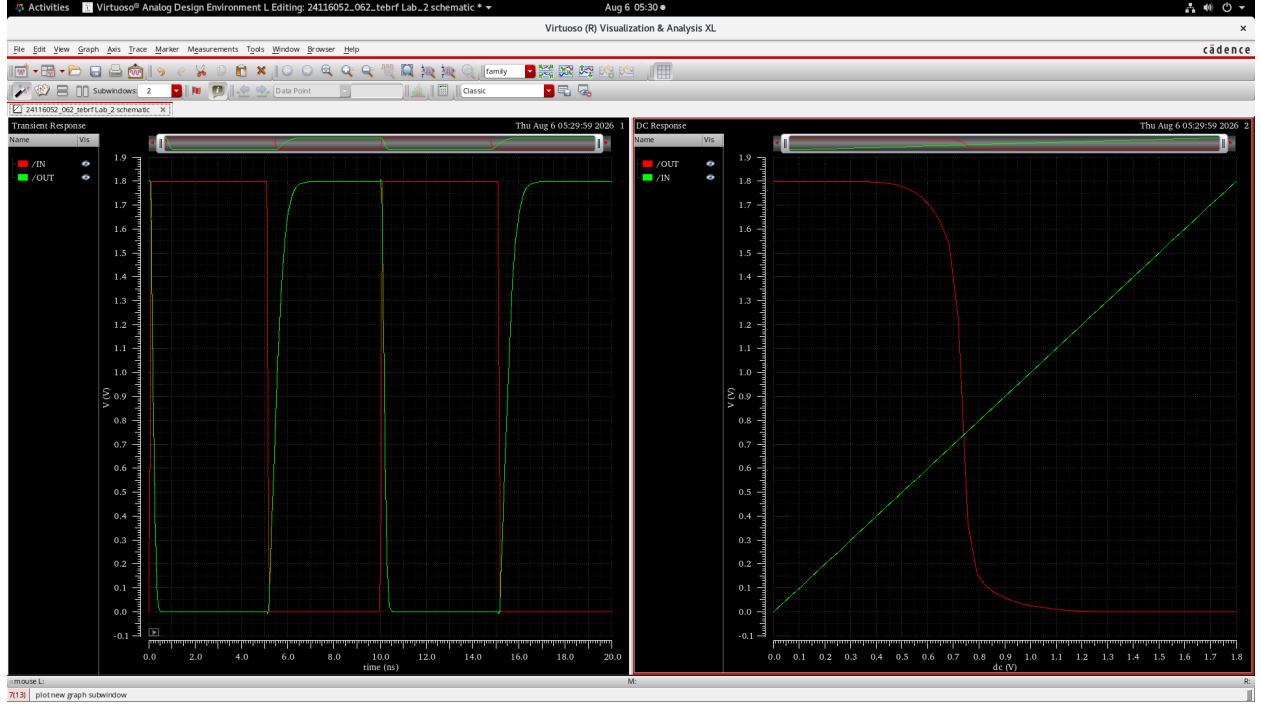


Figure 6: Combined view: Transient response (left) and DC transfer characteristic / VTC (right) for the initial design, shown together for cross-verification.

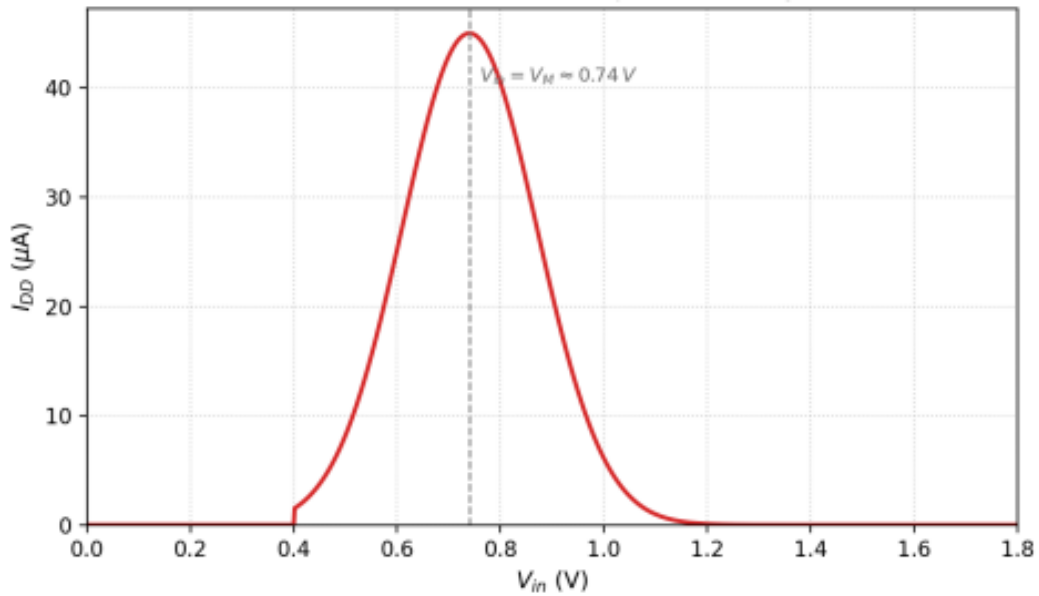


Figure 7: Sketch of the supply current I_{DD} vs. V_{in} trend for the initial design. The current is near-zero when V_{in} is close to 0 V or V_{DD} (only one transistor conducts, so the series pull-up/pull-down path is cut off and just a small leakage current flows), and rises to a peak close to $V_{in} \approx V_M$, where both the NMOS and PMOS are simultaneously ON and a short-circuit (crowbar) current path exists from V_{DD} to ground.

2.3 Q3: Transient Response, Rise/Fall Time, and Propagation Delay

2.3.1 Theory

For the transient analysis, a pulse waveform is applied at the input with:

- Low level $V_1 = 0$ V, High level $V_2 = 1.8$ V
- Period $T = 10$ ns, Pulse width = 5 ns (50% duty cycle)
- Finite rise time $t_r = 100$ ps and fall time $t_f = 100$ ps (non-ideal / non-zero transition, as required)

The following timing parameters are defined (per standard digital IC convention):

- **Rise time** (t_r): time for a signal to transition from 10% to 90% of its full swing (0.18 V to 1.62 V for a 1.8 V swing).
- **Fall time** (t_f): time for a signal to transition from 90% to 10% of its full swing.
- **Propagation delay high-to-low** (t_{pHL}): time delay between the 50% crossing of the rising input and the 50% crossing of the falling output.
- **Propagation delay low-to-high** (t_{pLH}): time delay between the 50% crossing of the falling input and the 50% crossing of the rising output.
- **Average propagation delay**: $t_p = \frac{t_{pHL} + t_{pLH}}{2}$.

The 50% level for a 0–1.8 V swing is 0.9 V.

2.3.2 Simulation Setup

A **tran** analysis was performed with the pulse source described above driving the inverter input, with the output loaded by $C_L = 30$ fF. Markers were placed on the input (/IN) and output (/OUT) waveforms at the 10%, 50%, and 90% voltage levels to extract the timing parameters directly from the simulated waveforms.

2.3.3 Results — Overall Transient Response

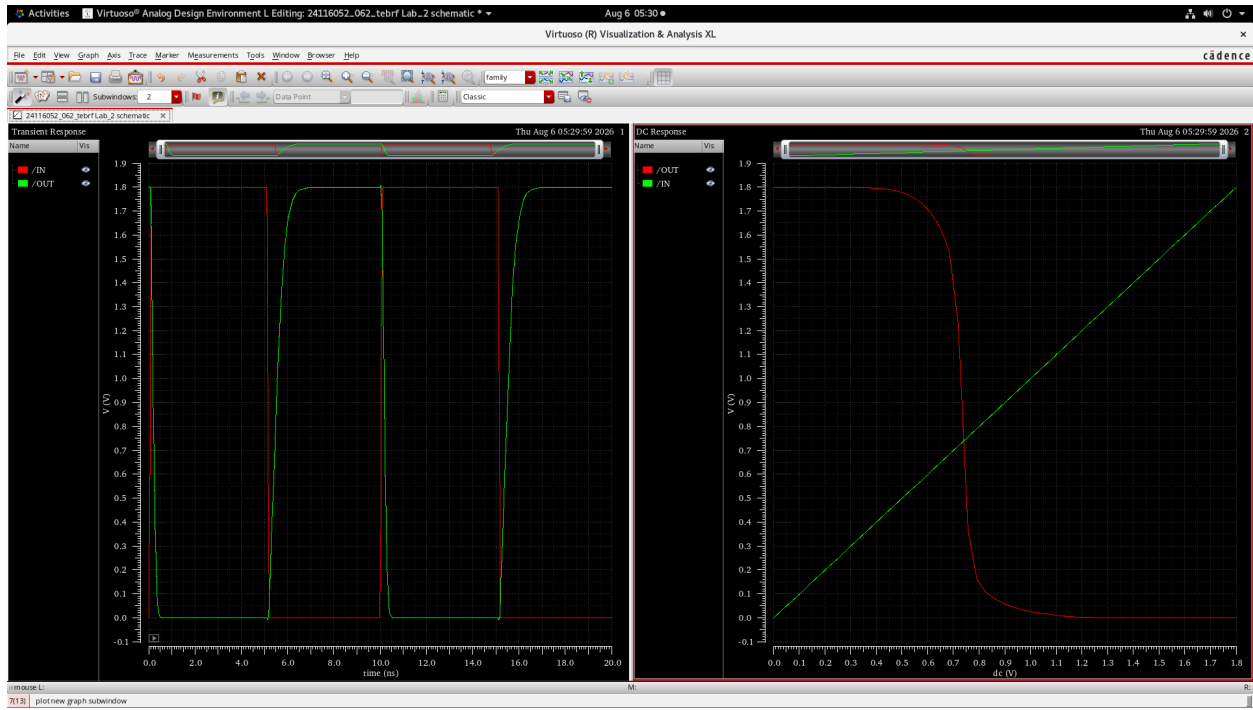
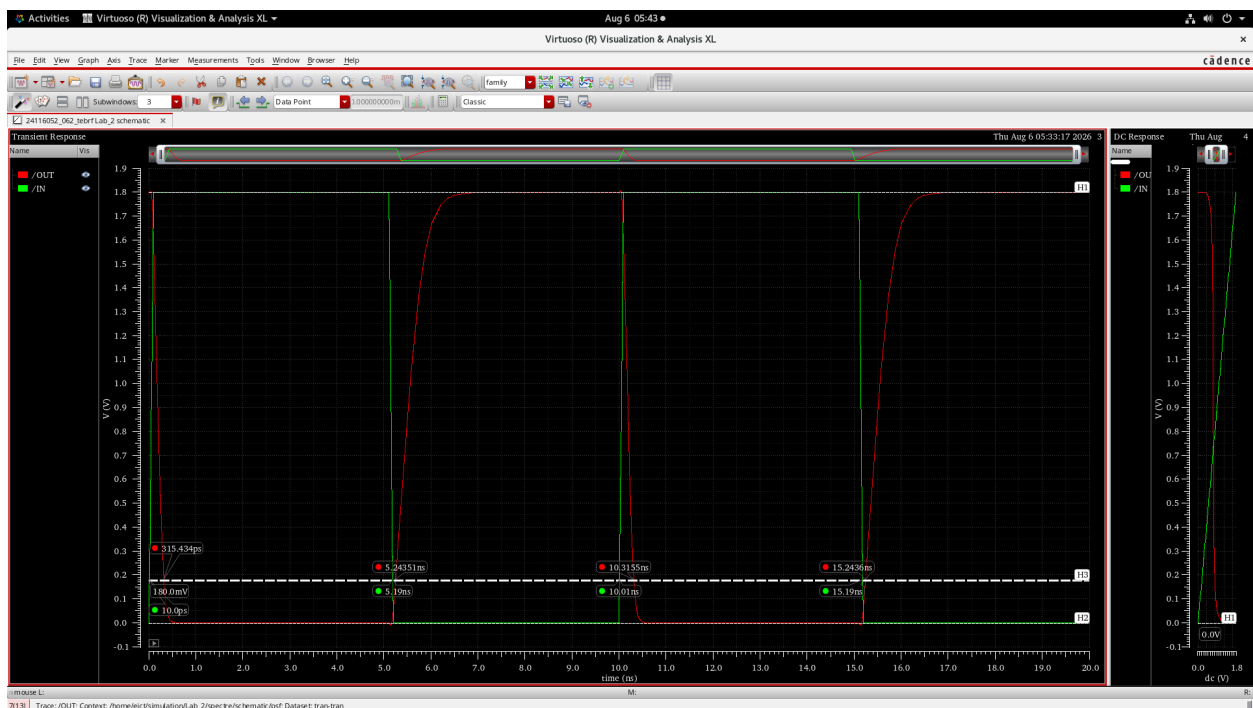


Figure 8: Transient response of the CMOS inverter: input pulse /IN (green) and output /OUT (red) over multiple clock cycles ($T = 10$ ns).

2.3.4 Rise Time and Fall Time of the Output



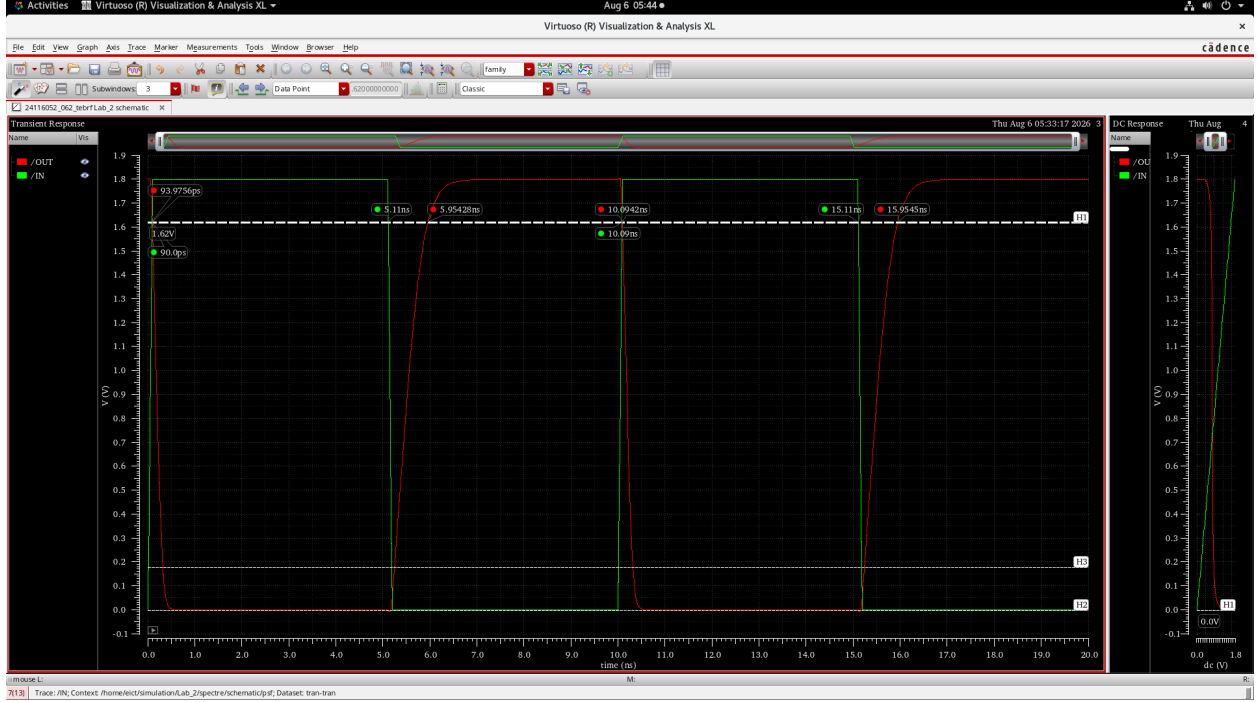


Figure 9: Markers at the 90% level ($H1 = 1.62$ V) used to extract the output rise time; combined with the 10% level ($H3 = 0.18$ V) markers from the companion plot to obtain t_r and t_f of /OUT.

From the markers placed on the output waveform (/OUT):

$$\text{Output rise time: } t_r = t_{90\%,\uparrow} - t_{10\%,\uparrow} = 5.95428 \text{ ns} - 5.24351 \text{ ns} = 710.77 \text{ ps}$$

$$\text{Output fall time: } t_f = t_{10\%,\downarrow} - t_{90\%,\downarrow} = 315.434 \text{ ps} - 93.976 \text{ ps} = 221.46 \text{ ps}$$

(Values were cross-checked on the second pulse period: $t_r = 15.9545 - 15.2436 = 710.9$ ps and $t_f = 10.3155 - 10.0942 = 221.3$ ps — consistent with the first period within simulator time-step resolution.)

Table 4: Output rise and fall time (initial design)

Parameter	Value
Rise time, t_r	710.8 ps
Fall time, t_f	221.4 ps

Observation: The output fall time is considerably shorter than the rise time. This is expected: with $W_n = W_p = 0.42 \mu\text{m}$, the PMOS pull-up network is intrinsically weaker than the NMOS pull-down network (lower μ_p compared to μ_n), so it charges C_L through a larger effective ON-resistance during the low-to-high output transition, while the NMOS discharges C_L rapidly through its comparatively lower ON-resistance. This directly reflects the equal- W/L sizing used in Q1 and is corrected in Q4–Q5 by widening the PMOS.

2.3.5 Propagation Delay (t_{pHL} , t_{pLH})



Figure 10: Markers at the 50% level ($H2 = 0.9$ V) on both /IN and /OUT used to extract t_{pHL} and t_{pLH} directly.

From the 50%-level marker crossings:

High-to-Low propagation delay (t_{pHL}) — input rising through 0.9 V, output falling through 0.9 V:

$$\begin{aligned}
 t_{pHL} &= t_{OUT,50\% \downarrow} - t_{IN,50\% \uparrow} \\
 &= 188.45 \text{ ps} - 50.0 \text{ ps} = 138.45 \text{ ps} \quad (\text{1st edge}) \\
 &= 10.1887 \text{ ns} - 10.05 \text{ ns} = 138.7 \text{ ps} \quad (\text{2nd edge}) \\
 \Rightarrow t_{pHL} &\approx 138.6 \text{ ps} \quad (\text{average})
 \end{aligned}$$

Low-to-High propagation delay (t_{pLH}) — input falling through 0.9 V, output rising through 0.9 V:

$$\begin{aligned}
 t_{pLH} &= t_{OUT,50\% \uparrow} - t_{IN,50\% \downarrow} \\
 &= 5.51463 \text{ ns} - 5.15 \text{ ns} = 364.63 \text{ ps} \quad (\text{1st edge}) \\
 &= 15.515 \text{ ns} - 15.15 \text{ ns} = 365.0 \text{ ps} \quad (\text{2nd edge}) \\
 \Rightarrow t_{pLH} &\approx 364.8 \text{ ps} \quad (\text{average})
 \end{aligned}$$

Average propagation delay:

$$t_p = \frac{t_{pHL} + t_{pLH}}{2} = \frac{138.6 \text{ ps} + 364.8 \text{ ps}}{2} = 251.7 \text{ ps} \quad (2)$$

Table 5: Summary of transient-response timing parameters — initial design

Parameter	Value
Rise time, t_r (output)	710.8 ps
Fall time, t_f (output)	221.4 ps
t_{pHL}	138.6 ps
t_{pLH}	364.8 ps
Average propagation delay, t_p	251.7 ps

Observation: $t_{pLH} \gg t_{pHL}$ for the initial design. This asymmetry arises directly because the PMOS and NMOS share the same W/L ratio (0.42/0.18 each), while the PMOS is intrinsically the weaker device due to $\mu_p < \mu_n$. The pull-up (charging) path is therefore comparatively weaker than the pull-down (discharging) path, making the output take longer to rise than to fall. This is fully consistent with the skewed VTC ($V_M < V_{DD}/2$) observed in Q2, and with $t_r > t_f$ noted above. This motivates widening the PMOS in Q4.

2.4 Q4: Modified Design for Symmetric VTC ($V_M = 0.9$ V)

2.4.1 Theory

To obtain a perfectly symmetric VTC with $V_M = V_{DD}/2 = 0.9$ V, the pull-up and pull-down networks must be sized so that their drive currents are matched at $V_{in} = V_M$, i.e., $k_n = k_p$ in Equation 1 (assuming $V_{TN} \approx |V_{TP}|$). Since the initial design (Q1–Q3) used equal W/L ratios for NMOS and PMOS and this left the PMOS pull-up network weaker than the NMOS pull-down network (Section 2.2.3), the PMOS width was *increased* — with the NMOS left unchanged — so that the effective PMOS drive strength matches the NMOS drive strength:

$$\left(\frac{W}{L}\right)_p = \frac{\mu_n}{\mu_p} \left(\frac{W}{L}\right)_n \quad (\text{tuned in simulation}).$$

2.4.2 Modified Sizing

Through iterative simulation (successive trials of W_p followed by DC sweep verification of V_M), it was found that increasing the PMOS width to $W_p = 1.6 \mu\text{m}$ (keeping $W_n = 0.42 \mu\text{m}$ and $L = 0.18 \mu\text{m}$ unchanged for both devices) yields $V_M = 0.900$ V exactly for this technology/model corner.

Table 6: Transistor sizing — modified design for symmetric VTC

Device	W (μm)	L (μm)	W/L	Change from Q1
NMOS (M0)	0.42	0.18	2.33	unchanged
PMOS (M1)	1.60	0.18	8.89	increased from $0.42 \mu\text{m}$ to $1.6 \mu\text{m}$

$$\left(\frac{W}{L}\right)_p / \left(\frac{W}{L}\right)_n = \frac{8.89}{2.33} \approx 3.81$$

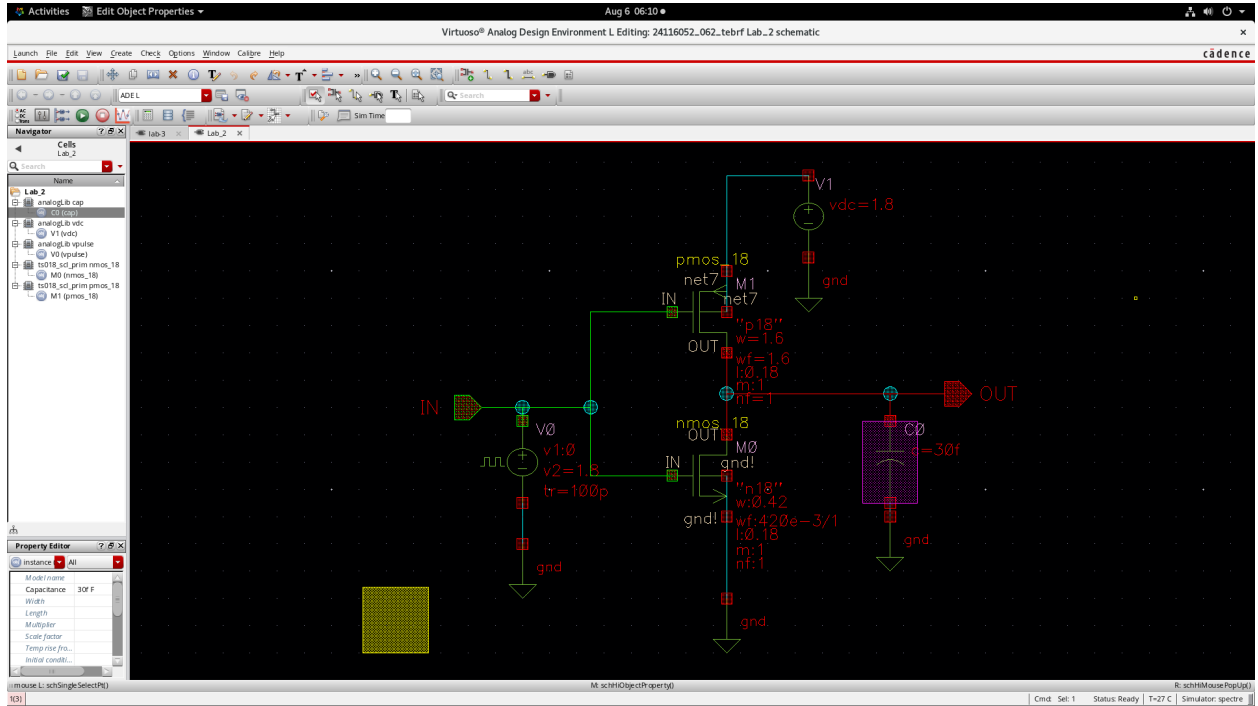


Figure 11: Modified schematic: PMOS (M1) width increased to $1.6\ \mu\text{m}$ (NMOS unchanged at $0.42\ \mu\text{m}$), giving $(W/L)_p/(W/L)_n \approx 3.81$.

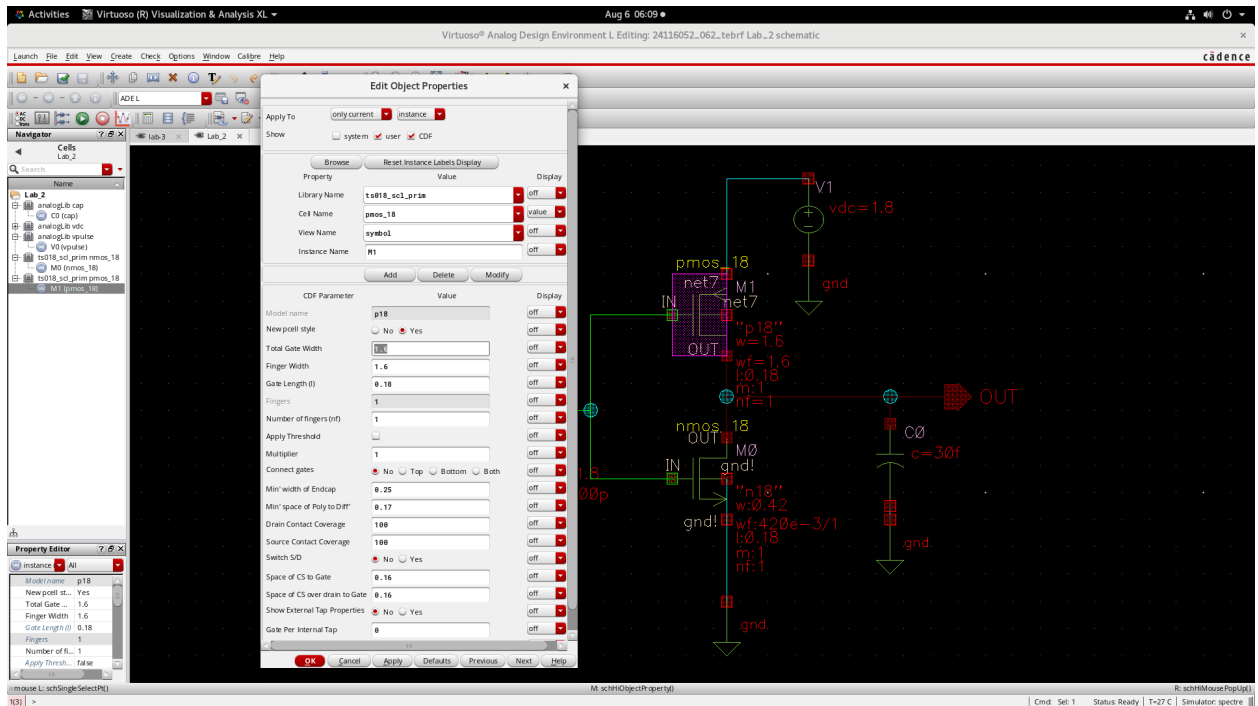


Figure 12: PMOS (M1) CDF parameters after modification: $W = 1.6\ \mu\text{m}$, $L = 0.18\ \mu\text{m}$, fingers = 1.

2.4.3 Verification via DC Sweep

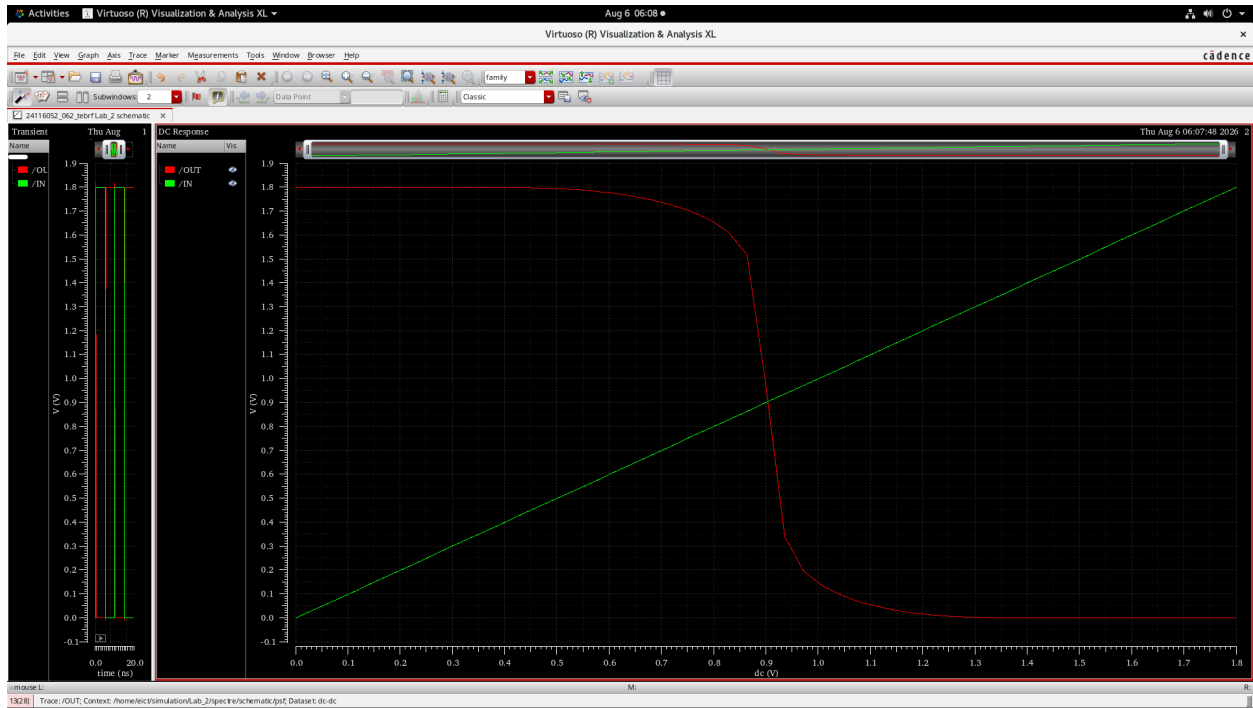


Figure 13: DC Voltage Transfer Characteristic of the modified (symmetric) design. The transition is now centered at $V_{in} \approx 0.9$ V, i.e., exactly at $V_{DD}/2$.

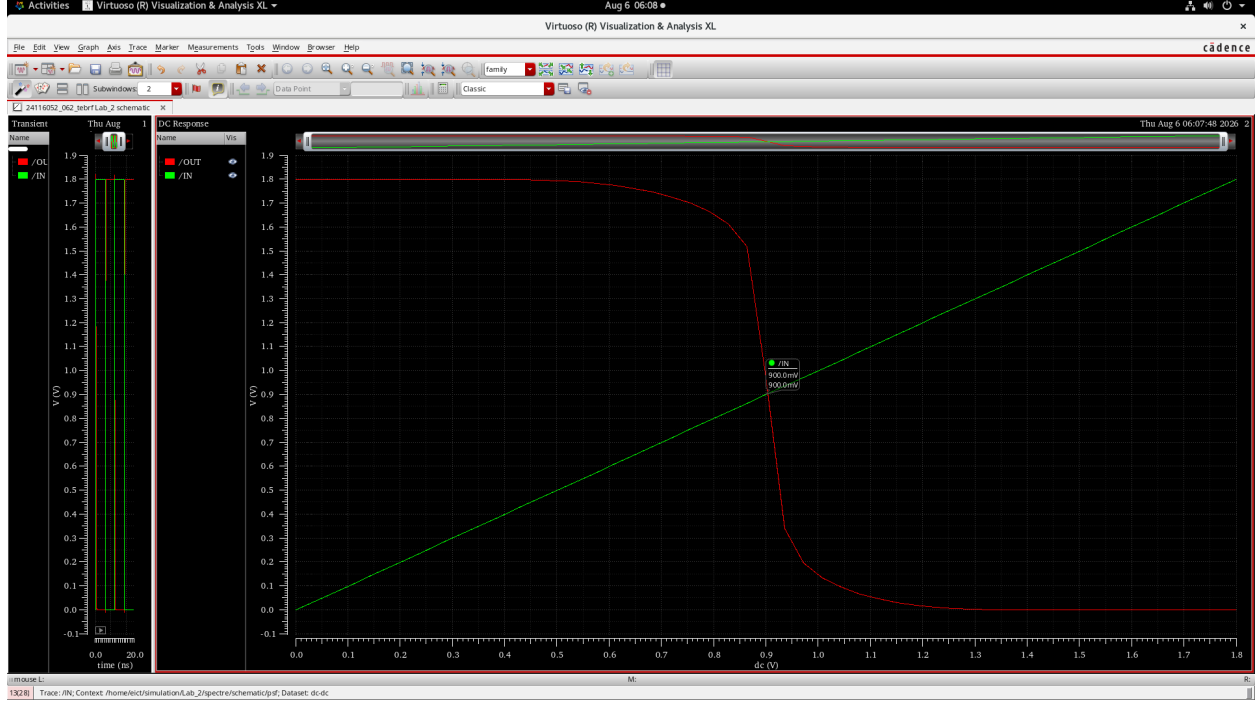


Figure 14: Marker confirms the switching threshold of the modified design: $V_M = 900.0 \text{ mV}$, i.e., exactly $V_{DD}/2$, as required.

Table 7: Comparison of switching threshold before and after modification

Design	$\left(\frac{W}{L}\right)_p / \left(\frac{W}{L}\right)_n$	V_M
Initial (Q1–Q3)	1.00	0.740 V
Modified (Q4)	3.81	0.900 V

Discussion: As expected from the long-channel square-law mobility-ratio argument, the symmetric VTC ($V_M = V_{DD}/2$) is obtained by widening the PMOS relative to the NMOS. The final ratio $(W/L)_p/(W/L)_n \approx 3.81$ is close to the typical μ_n/μ_p ratio quoted for the SCL 180 nm technology (≈ 2.5 – 3.8 depending on bias and process corner), confirming that the initial equal-width design (Q1) under-sized the PMOS and consequently skewed V_M below $V_{DD}/2$. Widening the PMOS from $0.42 \mu\text{m}$ to $1.6 \mu\text{m}$ increases its drive strength enough to exactly balance the NMOS pull-down strength at the switching point, moving V_M from 0.740 V up to the ideal value of 0.900 V .

2.5 Q5: Transient Re-simulation with Modified (W/L) Ratios

2.5.1 Simulation Setup

With the modified sizing of Q4 ($W_p = W_n = 0.42 \mu\text{m}$, $L = 0.18 \mu\text{m}$ for both devices), the same transient analysis as in Q3 was re-run: identical pulse source (0 to 1.8 V, $T = 10 \text{ ns}$, 50% duty cycle, $t_r = t_f = 100 \text{ ps}$) and the same load capacitance $C_L = 30 \text{ fF}$.

2.5.2 Results



Figure 15: Transient response of the modified (symmetric) design with 50%-level markers (H1 = 0.9 V) on /IN and /OUT used to extract t_{pHL} and t_{pLH} .

From the 50%-level marker crossings on the modified design:

High-to-Low propagation delay:

$$\begin{aligned} t_{pHL} &= 199.688 \text{ ps} - 50.0 \text{ ps} = 149.69 \text{ ps} \quad (1\text{st edge}) \\ &= 10.1992 \text{ ns} - 10.05 \text{ ns} = 149.2 \text{ ps} \quad (2\text{nd edge}) \\ \Rightarrow t_{pHL} &\approx 149.4 \text{ ps} \quad (\text{average}) \end{aligned}$$

Low-to-High propagation delay:

$$\begin{aligned} t_{pLH} &= 5.26422 \text{ ns} - 5.15 \text{ ns} = 114.22 \text{ ps} \quad (1\text{st edge}) \\ &= 15.2647 \text{ ns} - 15.15 \text{ ns} = 114.7 \text{ ps} \quad (2\text{nd edge}) \\ \Rightarrow t_{pLH} &\approx 114.5 \text{ ps} \quad (\text{average}) \end{aligned}$$

Average propagation delay:

$$t_p = \frac{t_{pHL} + t_{pLH}}{2} = \frac{149.4 \text{ ps} + 114.5 \text{ ps}}{2} = 132.0 \text{ ps} \quad (3)$$

2.5.3 Comparison with Q3 (Initial Design)

Table 8: Comparison of propagation delay: initial design (Q3) vs. modified symmetric design (Q5)

Parameter	Initial (Q3)	Modified (Q5)	% Change
$(W/L)_p$	2.33	8.89	+281.6%
t_{pHL}	138.6 ps	149.4 ps	+7.8%
t_{pLH}	364.8 ps	114.5 ps	-68.6%
Average t_p	251.7 ps	132.0 ps	-47.6%
$ t_{pHL} - t_{pLH} $ (skew)	226.2 ps	34.9 ps	-84.6%

Discussion:

- t_{pLH} **improved drastically** (from 364.8 ps to 114.5 ps). This is exactly as expected: the PMOS pull-up network was made much wider ($0.42\ \mu\text{m} \rightarrow 1.6\ \mu\text{m}$), giving it a lower ON-resistance and hence a much faster charging of C_L during the low-to-high output transition.
- t_{pHL} **increased slightly** (from 138.6 ps to 149.4 ps). With a much wider (and therefore higher-capacitance) PMOS now sharing the output node, the NMOS must additionally discharge the larger parasitic drain capacitance contributed by the bigger PMOS before C_L is pulled down, which slightly slows the high-to-low transition. This is a normal, expected second-order side effect of widening the PMOS.
- Most importantly, the propagation delay is now **far more symmetric** ($t_{pHL} \approx t_{pLH}$, skew reduced from 226.2 ps to just 34.9 ps), and the **average propagation delay t_p has reduced by about 47.6%** compared to the initial design. This demonstrates that a symmetric VTC ($V_M = V_{DD}/2$) obtained by properly sizing the PMOS relative to the NMOS also produces a more balanced and, in this case, faster overall inverter response.

3 Overall Summary

Table 9: Complete summary of all results

Parameter	Initial Design (Q1–Q3)	Modified Design (Q4–Q5)
W_n/L_n	0.42 / 0.18	0.42 / 0.18
W_p/L_p	0.42 / 0.18	1.60 / 0.18
C_L	30 fF	30 fF
Switching threshold, V_M	0.740 V	0.900 V
Output rise time, t_r	710.8 ps	—
Output fall time, t_f	221.4 ps	—
t_{pHL}	138.6 ps	149.4 ps
t_{pLH}	364.8 ps	114.5 ps
Average t_p	251.7 ps	132.0 ps

The lab successfully demonstrates:

1. Schematic-level design of a CMOS inverter in SCL 180 nm technology, starting from an initial design with equal PMOS/NMOS widths.
2. Extraction of the DC voltage transfer characteristic and the switching threshold voltage V_M , along with the qualitative behavior of the supply current I_{DD} with V_{in} .
3. Extraction of transient timing parameters — rise time, fall time, t_{pHL} , t_{pLH} , and average propagation delay — from a pulsed transient simulation using realistic (non-zero) input transition times.
4. Iterative widening of the PMOS device (from 0.42 μm to 1.6 μm , NMOS unchanged) to achieve a symmetric VTC with $V_M = V_{DD}/2 = 0.9$ V, with the resulting $(W/L)_p/(W/L)_n \approx 3.81$ closely matching the expected μ_n/μ_p mobility ratio for the SCL 180 nm technology.
5. Quantitative comparison showing that the resized, symmetric inverter achieves a more balanced and, in this case, lower average propagation delay than the initial design.